

Minh Le Bui

+1-(608)-598-8955 | minhbui@gmail.com | [linkedin.com/in/minhlembui](https://www.linkedin.com/in/minhlembui) | github.com/M1nhbui

EDUCATION

University of Wisconsin - Madison

Bachelor of Science, Computer Science & Data Science (GPA: 4.0/4.0)

Certificate, Economic Analytics

May. 2028

Madison, WI

- **Awards:** Dean's List - All Semesters
- **Leadership:** Vice President - Vietnamese International Student Association
- **Relevant Coursework:** Data Structures and Algorithms (A), Object-Oriented Programming (A), Distributed System (A), Computer Architecture (A), Data Science Programming (A), Operating System (A), Data Management (A), Statistical Modeling (A), Machine Learning, Database Management Systems

TECHNICAL SKILLS

Specialization: Software Engineering (Full-Stack), Data Engineering, Cloud Infrastructure, Machine Learning & LLMs

Programming Languages: Python 3, SQL, Java, C++, JavaScript, HTML, CSS, R

Databases: PostgreSQL, MySQL, MongoDB, Elasticsearch

Frameworks & Libraries: PySpark, PyTorch, LightGBM, XGBoost, Transformers, LangChain, FastAPI, Flask, SQLAlchemy, Pandas

Cloud & Tools: AWS, Firebase, Git, GitHub Actions, Docker, Linux, Conda, VMware Fusion, Tableau

EXPERIENCE

CMC OpenAI

AI Intern | Transformers, OCR

Jun. 2026 – Present

Hanoi, Vietnam

- Validated a **3,000-image** table dataset for fine-tuning **PaddleOCR-VL 1.6**, identifying **30%** mislabeled samples and systematic annotation errors before training.
- Analyzed **2,000** PaddleOCR-VL outputs to characterize model weaknesses on Vietnamese legislation documents, filtering a large data pool to high-value images that reduced training size while improving stability.

Thayer School of Engineering at Dartmouth – Advisor: Prof. Eugene Santos

Research Assistant | Llama-3.2-3B, LLM Fine-Tuning, Benchmarking, LangChain

Sep. 2025 – Present

Remote

- Fine-tuned **Llama-3.2-3B** (LoRA/QLoRA/full) to study hallucination, cutting cross-paraphrase answer variance $\sim 3\times$ (**49%→17%**) and hitting **100% consistency** on held-out facts; LoRA/QLoRA matched full fine-tuning at far fewer parameters.
- Compiled **40+** LLM hallucination and red-teaming benchmarks, standardizing metadata (scale, domain, method, limitations) to support **black-box and white-box evaluation** of **6** in-house models across factual accuracy and adversarial robustness.

VNPT-Media

Machine Learning Engineer Intern | XGBoost, Docker, FastAPI, PySpark

Jul. 2025 – Sep. 2025

Hanoi, Vietnam

- Optimized a **PySpark** feature engineering pipeline, processing **2M+** monthly user records to facilitate monthly retraining and ensure long-term model stability.
- Developed and deployed an **XGBoost** model to predict return likelihood of one-way locked users, achieving a final F1 score of **0.82 (14% improvement)** over the 0.72 Logistic Regression baseline; deployed internally via **Docker** and **FastAPI**.
- Performed systematic experimentation with feature selection and Bayesian **hyperparameter tuning**, leading to **+5%** in predictive performance; leveraged SHAP (Explainable AI) to explain model outputs and support non-technical stakeholders.

AI Research Center - Hanoi University of Science and Technology

Research Intern | RAG, FAISS, PyTorch, Transformers, VLMs, CLIP

May. 2025 – Jul. 2025

Hanoi, Vietnam

- Prototyped a **Retrieval-Augmented Generation (RAG)** pipeline using **FAISS**, **PyTorch**, and **Transformers** for PET/CT medical scans, delivering a foundational retrieval framework to accelerate future Vision-Language Model testing.

Biokind Analytics

Data Engineer & Data Analyst | Pandas, GeoPandas, Matplotlib, Tableau

Sep. 2024 – May. 2025

Madison, WI

- Automated data collection and cleaning for 15+ counties via FIPS codes, reducing manual processing time by **40%**.
- Analyzed **12GB+** of historical data (6 years) using **Pandas**, directly informing a partner bank's donation strategy.
- Created interactive **Tableau** and **GeoPandas** dashboards for stakeholders to plan upcoming donation campaigns.

TECHNICAL PROJECTS

Real-Time News Sentiment Dashboard | Transformers, GitHub Actions, Streamlit, PostgreSQL, AWS, Docker

- Building a scalable sentiment analysis pipeline by fine-tuning a Hugging Face **DistilBERT** model on **AWS SageMaker**, with planned deployment via dockerized **AWS Lambda** + API Gateway for **sub-second** news classification.
- Developed a fully automated ETL pipeline using **NewsAPI** and **Pandas**, with CI/CD orchestration via **GitHub Actions**, ingesting, cleaning, and labeling **30+ political articles/day** with an end-to-end latency of **<3 minutes** and 100% uptime.
- Stored data in **PostgreSQL** (AWS RDS) using SQLAlchemy, achieving **93% faster queries** (3s to 200ms).
- Deployed Streamlit dashboard with auto-refresh visualizations (Altair), containerized in **Docker** with 24/7 uptime.

AWARDS & ACHIEVEMENTS

ICPC North Central North America (NCNA) Regional 2025 – Placed 14th out of 86 teams

Vietnam Science and Engineering Fair (ViSEF) – 4th prize (System Software)